Strange Compass Transformers: Geometric Navigation in Semantic Embedding Space

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Abstract

Current approaches to semantic relational computation in neural language models suffer from two fundamental limitations: autoregressive models require expensive sequential computation for each transformation, while similarity-based approaches can only find nearby concepts, lacking a grammar of arbitrary relational navigation. These methods lack the ability to perform precise, interpretable operations on meaning.

We introduce Strange Compass Transformers (SCTs), a novel neural architecture that performs explicit geometric transformations in semantic embedding space through learned 2D rotation planes. Unlike traditional language models that generate text autoregressively, SCTs operate as semantic actuators, accepting structured inputs specifying angular transformations (0°–360°), magnitude scaling (0.0–1.0), and contextual domains to navigate meaning systematically.

A set of innovations underpin SCTs. Pivotally, the core innovation is ontological, inspired by the work of Hofstadter and Sander in their landmark *Surfaces and Essences*; the thesis of Hofstadter's work is that analogical computation, as it occurs in biological brains, is the core of thinking. The Strange Compass Transformer takes its "strange" namesake from this Hofstadterian "strange loop" perspective; it also aims to experimentally test the empirical utility of deeply integrating analogical navigation as a primitive into ML-based systems. The remaining innovations of the SCT, including the compass metaphor and the unique multi-plane architecture, are contributions by the author as solutions to the polysemy problem and the distinctive training challenges of such a model architecture.

Instrumentally, SCT operates by projecting high-dimensional embeddings onto learnable 2D manifolds where clean geometric rotations occur, then lifting the transformed representation back

to embedding space. We have developed distinct **Shadow Plane Adapters**, per-context learned projections that absorb the geometric mismatch between pre-trained embedding structure and rotation semantics, a mitigation strategy with regards to the polysemy problem that arises when words participate in multiple semantic contexts.

The architecture aims to achieve semantic navigation with **~131M parameters** through an **8-head compass system** (4 cardinal + 4 intercardinal directions) with **uneven Q/K/V attention splits** that emit continuous proximity signals. Through end-to-end learning that jointly optimizes shadow projections, alignment transforms, and geometric operations, SCTs learn to map angular operations (180° for opposites, 90° for analogies) to semantic relationships while maintaining mathematical properties like involution and magnitude monotonicity.

By externalizing analogical structure as a queryable geometric primitive, SCT provides reusable relational infrastructure for downstream reasoning systems, amortizing the cost of learning semantic relationships across tasks. Please see Section 2 for a thorough exposition of the underlying cognitive science by Hofstadter, Sander, and Slotnick respectively.

Keywords: geometric deep learning, semantic transformations, 2D rotation planes, shadow plane adapters, proximity-gated attention, compass architecture, analogical reasoning, conceptual compression

1. Introduction

Current approaches to semantic manipulation in neural language models suffer from two fundamental limitations: autoregressive models require expensive sequential computation for each transformation, while similarity-based approaches can only find nearby concepts rather than systematically navigate semantic space. These methods lack the ability to perform precise, interpretable geometric operations on meaning.

We propose Strange Compass Transformers (SCTs), a new class of models that treat semantic transformation as geometric navigation in learned 2D planes. Rather than attempting rotation in intractable high-dimensional space, SCTs project embeddings onto context-specific 2D manifolds where clean algebraic rotations can occur, then lift the results back to full embedding space. This approach provides **steerable exploration** for reasoning systems through precise angular and magnitude parameters.

A fundamental challenge in this design space is the **polysemy problem**. When words like "cold" participate in multiple semantic contexts (physics: temperature; medicine: symptoms; psychology: personality), training creates gradient conflicts as each context pulls the shared embedding in different directions. We address this through **Shadow Plane Adapters**, per-context learned projections that absorb context-specific geometric distortion, isolating adaptation from the shared embedding space and preserving interpretable rotation semantics.

The theoretical warrant for the entire enterprise rests on a thesis drawn from cognitive science, principally Hofstadter and Sander's *Surfaces and Essences* [3] and Slotnick's *The Cognitive Neuroscience of Memory* [11]: that analogical reasoning constitutes a form of compressed latent computation, and that SCT operationalizes this compression as navigable geometry. Section 2 develops this argument in detail.

Beyond interpretability, the most concrete motivation for SCT is as a deterministic steering primitive that autonomous cognitive architectures can *call*: hypergraph memory traversal, forced angular reorientation to break a stalled chain of thought, dream-phase memory consolidation, and the deliberate broadening of database search. To be precise about what this is and is not: SCT is not an activation-level intervention applied inside a language model's forward pass, and it does not condition token-probability distributions during decoding. It is a standalone, non-autoregressive relational transform that an external system queries on demand; the "steering" it provides is the querying of a relational geometry, not the rotation of activations during generation. This position paper grounds the architecture and specifies the mathematics; a companion empirical paper will have room to lead with these applications.

1.1 Contributions

1. **Philosophical Grounding**: We make the paper's central hypothesis explicit (analogy as compressed latent computation), grounding each stated assumption in cited cognitive science and clearly marking the paper's own theoretical contributions.
2. **Cognitive Science Integration**: We integrate Hofstadter and Sander's account of analogy as constitutive of cognition, their "conceptual skeleton" mechanism of structural compression, and Slotnick's neuroscience evidence for the speed advantage of pattern-based retrieval.
3. **Shadow Plane Architecture**: Per-context adapter layers that solve the polysemy problem by factorizing adaptation between learned shadow projections and interpretable rotation planes.
4. **2D Projection Solution**: We formalize the projection-rotation-lift pipeline that makes high-dimensional semantic rotation tractable through learned 2D manifolds.
5. **Proximity Reward Architecture**: A novel reward function that trains attention heads to emit **continuous angular proximity signals** rather than binary activations, enabling natural interpolation via V-signal weighted averaging.
6. **Learning Rate Hierarchy**: A principled gradient of learning rates across architectural components, with empirical tuning guidance for production deployment.
7. **Practical Implementation**: A ~131M parameter model with 120k vocabulary suitable for domain-specific terminology, trainable on consumer GPU hardware.

2. Philosophical Foundations

2.1 The Central Hypothesis

This paper rests on a hypothesis to be tested by building an SCT and publishing findings in the complete SCT paper (due later this year).

Assumption 1 *(grounded in cognitive science)*: Analogy is constitutive of cognition. It operates by extracting reusable relational structures, "conceptual skeletons," that compress experiential knowledge into forms callable without recapitulating the experiences that produced them.

Assumption 2 *(the paper's interpretive contribution)*: If we characterize this cognitive operation in computational terms, it functions as information-theoretic compression: encoding relational patterns once, then amortizing retrieval across future encounters. We call this **compressed latent computation**. Compressed, because the analogical solution space has been precomputed (mapped) by centuries of human thought and action, then compressed into representationally dense analogical relations. It is this process that discards the surface detail in favor of transferable relational cores, that we attempt to distill into word symbol relations using a spatial navigation metaphor as access mechanism. Latent, because the structures' relational value and precomputed intelligence sit disinterestedly awaiting only the relational activation pathway that maps an input to an output. Computational, because they produce determinate outputs (adaptive responses) from structured inputs (novel situations).

Prediction: An ML model that internalizes this compressed relational structure and acts as a queryable geometric primitive (SCT) can provide the same amortized benefit to artificial agents that analogical cognition provides to biological ones.

We note clearly: Assumption 1 is well-supported by the cognitive science literature, as we document below. Assumption 2 is our own theoretical contribution, a deliberate computational reframing of cognitive phenomena.

The prediction is our test, if SCT performs as predicted, it follows that it can be harnessed for the directed steering of powerful inferential systems.

We do not claim certainty; we claim a hypothesis worth testing with experimental rigor.

On what is fixed and what is not. The commitment of this work is to the *realization* of a Strange Compass Transformer — a trainable primitive that externalizes analogical structure as queryable geometry — and not to any particular blueprint for achieving it. The architecture specified in this paper (shadow plane adapters, the eight-head compass, the projection–rotation–lift pipeline, the alignment variants of Section 3.4) is our leading conception of how to build that primitive: the design we currently judge most likely to succeed and most interpretable in its failure modes. It is the starting point of an empirical search, not a fixed object to be defended. Where implementation and training expose flaws — a projection that discards too much structure, an alignment family too weak or too unconstrained, a compass topology that misencodes some relation, a proximity signal that will not train against its intended target — we will revise the architecture accordingly, including toward designs materially different from the one presented here. What persists across such revisions is the thesis under test and the goal it serves; the specific geometry is provisional. We state this as a methodological commitment rather than a hedge: the worth of the program lies in subjecting the idea to honest construction, and honest construction means allowing the artifact, not the author's prior conviction, to decide which mechanisms survive.

2.2 Analogy as the Mechanism of Thought

2.2.1 The Constitutive Claim

The strongest version of the case for analogy's cognitive centrality comes from Hofstadter and Sander [3], who argue that analogy-making and categorization are "just two names of the same phenomenon." Not two cognitive operations that occasionally collaborate, but a single process viewed from different angles. Their central thesis is unambiguous: "analogy-making is the mechanism at the basis of all thought" [3].

This is stronger than treating analogy as cognitive *infrastructure*, a useful tool that solves the problem of rapid response. Hofstadter and Sander argue it is the cognitive *process itself*: "the spotting of analogies pervades every moment of our thought, thus constituting thought's core" [3]. On their account, we do not *use* analogies to think; analogical pattern-matching *is* what thinking consists of.

The claim is grounded in three interlocking observations.

Ubiquity and frequency. Hofstadter and Sander argue that analogical mappings do not occur occasionally, during moments of creative insight or deliberate problem-solving, but continuously and at extraordinary frequency: "numerous times every second," such that we "swim nonstop in an ocean of small, medium-sized, and large analogies" [3]. This is not a claim about special cognition; it is a claim about *default* cognition. Even "the simplest and plainest of words and phrases that we come out with" arise from "rapidly, unconsciously made analogies" [3].

Automaticity and unconsciousness. The process operates largely below conscious threshold. Hofstadter and Sander document how everyday perception relies on instant analogical mapping: seeing someone unfold a map, scratch an elbow, or yawn triggers immediate recognition because "we, too, have" performed these actions many times [3]. These mappings are "not necessarily deeply insightful" in content, yet are "deep" in function because they lie "at the roots of our understanding of other beings" [3]. Past concepts are "selectively mobilized instant by instant, and nearly always without any awareness on our part" [3]. What Hofstadter and Sander call "naïve analogies" lead "directly to conclusions without any consideration of other options" and without the analogy being consciously felt [3]. This is a description of a cognitive system optimized for speed at the expense of deliberation.

Foundational necessity. Their strongest formulation: "No thought can be formed that isn't informed by the past; or, more precisely, we think only thanks to analogies that link our present to our past" [3]. This positions analogy not as one reasoning strategy among several, but as the substrate upon which all reasoning depends. Without it, the world would appear as it does to a newborn, where "each new concept has to be acquired from scratch" [3].

2.2.2 The Compression Mechanism

What does this continuous analogical process actually *do*? Hofstadter and Sander's answer centers on a mechanism they call the **conceptual skeleton**: an abstract relational pattern that survives the stripping away of surface detail.

"The very essence of an analogy is that it maps some mental structure onto another mental structure" [3]. What transfers between situations is not raw perceptual content but a distilled relational core. In their recurring example, multiple superficially different episodes share the skeleton "an unfortunate incident brought about by the very act of trying to avoid it" [3]. This is a causal pattern compact enough to store, general enough to match across domains, and specific enough to guide adaptive response when encountered again.

This skeleton can be, in their words, "clear as a bell" [3]. Not a subjective impression but a recoverable structural commonality that multiple observers converge on. The cognitive payoff is direct: by "seeing the new in terms of the old and familiar," organisms reuse prior knowledge with "only a slight cognitive cost" [3]. The alternative, perceiving each situation afresh, would require cognitive resources that biological systems under time pressure cannot afford.

The mechanism operates across every scale of cognition. In scientific discovery, Hofstadter and Sander document how "nearly all the great discoveries of the last three centuries" in physics have a "crucial, decisive analogy" at their heart [3]: gravitational potential understood as analogous to a hill (Lagrange & Laplace, 1770), electric potential recast via gravitational potential (Poisson, 1811), magnetic potential mapped through electric potential (Maxwell, 1855), and so on through de Broglie, Heisenberg, Schrödinger, and Dirac. In each case, the mechanism is the same: importing a proven conceptual framework, a skeleton, into a new domain where it organizes otherwise intractable phenomena.

In everyday cognition, the same mechanism operates at smaller scale but higher frequency. The concept “grocery store” carries within it relational knowledge (where bananas are likely found, how aisles are organized) that was “acquired through the making of analogies and, when needed, it is triggered by analogy” [3]. What Hofstadter and Sander playfully term a “bananalogy” lets a person navigate an unfamiliar grocery store, “even one in a foreign country” [3], by reusing structural knowledge extracted from prior encounters. The skeleton (spatial layout expectations) transfers; the surfaces (specific store, specific country) are discarded.

2.2.3 Lossy but Functional

Critically, this compression is lossy. Hofstadter and Sander document how “naïve analogies” can be misleading, sometimes yielding “absurd conclusions or complete dead ends” [3]. They liken such analogies to “skiers who sail with grace down well-groomed slopes but who are utterly lost in powder” [3]. These are structures that work smoothly within their domain of applicability but fail when extended beyond it. A child who understands division only through the analogy of “sharing candies” will struggle with cases where the quotient exceeds the dividend [3]. A four-year-old watching his father shave may conclude that shaving cream *dissolves* stubble, a coherent inference from the familiar schema of “sugar dissolving in water,” but mechanistically wrong [3].

The lossiness is not a defect; it is the *source* of the speed. Complete causal models are precise but expensive. Compressed skeletons are approximate but cheap. Organisms that traded precision for wall-clock survival left more descendants than those that reasoned from first principles while the predator closed the distance.

Hofstadter and Sander note that naïve structures often “persist and coexist with more sophisticated knowledge rather than being replaced” [3]. The child’s concept of division-as-sharing does not disappear when formal division is learned; it remains as an available (and often useful) default. This coexistence means cognition maintains a portfolio of compressed representations at varying levels of fidelity, deploying whichever skeleton the current situation activates first. The system is optimized for expected utility across encounters, not for correctness on any single encounter.

2.3 From Cognition to Computation

2.3.1 The Interpretive Move

Section 2.2 documents what cognitive science has established: analogy operates continuously, compresses relational structure into reusable skeletons, and does so automatically and below conscious awareness. We now make the interpretive move that is this paper's own contribution.

We propose that the cognitive process Hofstadter and Sander describe can be usefully characterized as **compressed latent computation**:

- **Compressed**: Relational structure is encoded with surface detail discarded, preserving only the transferable skeleton. This is information-theoretic compression, reducing the dimensionality of experiential knowledge to its task-relevant relational core.
- **Latent**: The process operates "nearly always without any awareness on our part" [3]. The computations are not consciously accessed or directed; they run as background infrastructure beneath deliberate reasoning.
- **Computation**: Structured inputs (novel situations) produce determinate outputs (adaptive responses, category assignments, retrieved episodes) through systematic operations (pattern matching, structural alignment, skeleton extraction). Whether or not this constitutes "computation" in a strong Church-Turing sense is a philosophical question we do not need to resolve. We require only the weaker claim: the process is *systematic enough to be operationalized.* That is, the input-output mapping can be approximated by a learnable function.

We do not claim that "thinking is computation" as a metaphysical thesis. We claim that the specific cognitive operation Hofstadter and Sander describe, extracting reusable relational skeletons and matching them to novel situations, is sufficiently systematic that its *functional signature* can be captured by a computational architecture. SCT is our attempt to build that architecture.

2.3.2 Neuroscientific Evidence for Speed

Slotnick's work on memory retrieval provides modest but relevant support for the speed advantage of pattern-based processing [11]. Event-related potential (ERP) studies show that familiarity-based recognition signals (mid-frontal old-new effect) appear at 300–500ms, while recollection-based signals (left-parietal old-new effect) appear later at 500–800ms [11]. Tasks relying on implicit memory produce "much more rapid response times," and repetition priming yields "more fluent or efficient processing" even without conscious recollection [11].

Slotnick does not discuss analogy per se. But the evidence demonstrates that pattern-based retrieval, structurally similar to the analogical matching Hofstadter and Sander describe, is measurably faster than deliberative recall. The neural substrate appears to be optimized for exactly the kind of compressed relational lookup that SCT attempts to formalize: rapid, pattern-driven, operating below the threshold of explicit awareness.

Additionally, Slotnick treats pattern completion as a key process in memory retrieval, with theta-gamma alignment (theta-wave troughs and gamma-wave peaks aligning during retrieval) reflecting structured cross-frequency coupling during recall [11]. This structured alignment during retrieval parallels, at the neural level, the concept-activation-based reminding that Hofstadter and Sander describe at the cognitive level. Together, they suggest that the brain's memory hardware is organized around compressed structural matching as a fundamental operation.

2.4 Concept Growth and the Urban Metaphor

Throughout *Surfaces and Essences*, Hofstadter and Sander describe concept development through a striking spatial metaphor: a concept begins as something like a small village, a narrow, nascent core of meaning built from early encounters. Through repeated analogical extension, it grows into a town, then a city, developing arteries and connections to neighboring concepts. New usages build what Hofstadter and Sander call "a blurry mental cloud of potential usages that are analogous to it" [3], and the concept expands outward as speakers explore "the fringes of these expanding categories" [3], reinforced or pruned by social feedback.

This metaphor is more than decorative. It maps onto what we know about neural development: dendritic arbors grow connections, synaptic strengths consolidate through repeated activation, and network topology reflects the history of experiential encounters. The "old city core" of a concept, its earliest, most prototypical meaning, remains structurally central even as suburbs of extended usage spread outward. Hofstadter and Sander note that concepts have "invisible roots" that remain "structured in that manner" even after the original analogical acts that built them are forgotten [3].

The urban metaphor also illuminates a critical property of conceptual structure: it is *spatial*. Hofstadter, as has been observed[3], constructs his argument about analogy almost entirely through spatialized language: centers and peripheries, cores and outskirts, nearness and remoteness of associations, travel from concrete to abstract. He speaks of conceptual "neighborhoods," of meanings that are "close" or "remote," of how categories have "fringes" that shade off into adjacent territory. This is the language of someone already working within an implicit topology of meaning, even while arguing that such topology resists formalization.

We wish to be precise about the relationship between Hofstadter's cognitive account and SCT's architecture. Hofstadter and Sander provide *inspiration*, not *isomorphism*. Their urban metaphor for concept growth describes a developmental process, how meanings expand and connect over a lifetime of use, while SCT is a *navigation* model that operates on a snapshot of the resulting structure. SCT does not grow concepts; it queries the relational geometry that concept growth has produced. The village-to-city trajectory describes how training data comes to encode the structure it encodes; SCT's rotation planes describe how that structure can be traversed once encoded.

The architectural debt to Hofstadter is real but specific: we take from him the conviction that relational structure between concepts is *systematic enough to navigate*, that it has directional and metric properties (things can be "closer" or "further" in meaning, and the direction matters), and that this structure was built by, and for, analogical reasoning. SCT is a unique operationalization of a semantic navigation model. It takes its philosophical warrant from Hofstadter and Sander's cognitive program while making its own engineering commitments about how to implement that navigation geometrically.

2.5 SCT as Reusable Analogical Infrastructure

Strange Compass Transformers attempt to externalize the cognitive infrastructure described above into a reusable computational primitive.

The hypothesis is direct: if analogical mappings helped humans survive by compressing relational knowledge into efficiently callable form, then equivalent compression could help artificial agents solve problems using less wall time.

SCT is not generalization itself. It is **one of the gears underneath a successful generalization system**: a mechanism that, given a concept and a relational query (opposite? analogy? complement?), returns a structurally appropriate response without requiring the downstream system to learn these relations from scratch.

Consider an agent reasoning about a novel domain. Without analogical infrastructure, it must learn that "success opposes failure," "heat relates to temperature," "buying complements selling," each relation acquired through costly examples. With SCT, these relations are pre-encoded and queryable: rotate by 180° for opposition, 90° for orthogonal relation, magnitude for degree of transformation.

Hofstadter and Sander's own account of cognitive reuse supports this framing directly. They describe how relational structures are "acquired through the making of analogies and, when needed, triggered by analogy" [3], built once through experience, then retrieved and applied whenever a structurally similar situation arises. Children build expanding webs of analogical usage that become invisible infrastructure: the "mommy"→"mother" conceptual link, the grocery-store-layout expectations, the causal skeletons that let us predict what happens when we drop fragile things. SCT proposes to externalize this same pattern: pre-encode relational structure so downstream systems can trigger it without building it.

Hofstadter and Sander explicitly engage with the AI implications of their account. They contrast an adult human, who "perceived the world using categorization through analogy," with a computer that "had no such mechanism," concluding the result is "lopsidedness… greatly in favor of humans" [3]. They treat analogical categorization as so fundamental that a machine lacking it would be "severely limited" [3], and they reference "Computer Models of the Fundamental Mechanisms of Thought" as a serious research direction [3]. SCT can be read as one answer to the challenge they pose: if machines without analogical structure are impoverished, then building queryable analogical infrastructure is not an optional enhancement but a structural necessity.

The value proposition is **amortized computation**. Train once on collective human analogical structure; deploy repeatedly across tasks that benefit from relational reasoning. The training cost is paid once; the inference benefit compounds across uses.

This positions SCT not as a standalone system but as **infrastructure**, analogous to how attention mechanisms, layer normalization, and residual connections serve as reusable primitives across diverse architectures. A reasoning system equipped with SCT can query relational structure without learning it, focusing its capacity on task-specific challenges.

2.6 On Objectivity and Validation

We make no claims about whether SCT's learned semantic structure corresponds to mind-independent reality. Such claims are epistemologically fraught and unnecessary for our purposes.

Instead, we adopt a pragmatic stance: semantic mappings have varying degrees of **utility**, measured by whether agents who navigate by them achieve desired outcomes. An SCT is "accurate" to the extent that its angular semantics compress relational structures that prove useful when applied.

This utility is transpersonal. A single user's success or failure is noise; aggregate outcomes across many navigations constitute signal. The "objectivity" of an SCT is thus a fluid scalar: the degree to which its mappings, when followed, yield outcomes better than chance across a population of users and use cases.

Hofstadter and Sander provide support for this pragmatic framing. They argue that analogy can be objective: "What could be more objective than these analogies?" they ask, citing examples like sound waves and ripples, lungs and gills, bullet and arrow [3]. They also acknowledge that analogy "comes in degrees of obviousness/strength" depending on how clearly the underlying essence is shared [3]. When the common core is "clear," the analogy is not merely personal but something "everyone" can see [3]. This is precisely the spectrum SCT's angular semantics inhabit: some rotations will produce universally recognized relationships ($180°$ on "hot" $\longrightarrow$ "cold" in physics), while others will be weaker or more context-dependent.

The relational structures encoded in human language are not arbitrary. They are **compressed mappings against experienced reality**, refined over millennia of collective use. Whether this refinement reflects alignment with physical law, convergent cognitive architecture, or robust cultural consensus is a question we leave to philosophers of language and mind. For engineering purposes, we require only that:

1. Some mappings yield better outcomes than others
2. This difference is measurable through aggregate use
3. Feedback loops can improve alignment over time

An SCT trained on human linguistic consensus inherits whatever validity that consensus possesses. Its field lines point not toward Platonic truth but toward **collective experiential structure**: what communities found useful for navigating the world. Deployment feedback can refine these field lines toward broader consensus, much as collaborative systems harness collective judgment to solve problems no individual annotation could address.

In this view, a trained SCT is a **cultural-cognitive artifact**: a crystallized snapshot of relational semantics as understood by a community at a time, encoding what that community believed about the directional structure of meaning. Different communities, across languages, legal traditions, and historical periods, may produce different SCTs, each internally consistent, each subject to empirical validation through use. We note that this last claim, cultural variation in analogical structure, is our own extrapolation; the cognitive science literature we cite does not directly address systematic cross-cultural differences in analogical mapping [3].

2.7 Implications for Artificial General Intelligence

If SCTs work as hypothesized, providing queryable analogical structure that transfers across domains, they may contribute meaningfully to the broader quest for general artificial intelligence. Not as complete solutions, but as **cognitive primitives** that reduce the learning burden for systems that must reason relationally.

The modest computational requirements of SCT training merit attention here. Current hardware trends suggest that full SCT pre-training may soon complete in a single day on accessible hardware, with fine-tuning requiring far less. If analogical structure can be trained and updated rapidly, the possibility emerges of **dynamic analogical adaptation**: SCTs that evolve with their user communities, refining field lines in response to aggregate feedback.

This is speculative but grounded. The architectural choices we describe (shadow plane adapters, learning rate hierarchies, continuous angular interpolation) are designed with such adaptability in mind. Whether the speculation proves warranted awaits empirical validation.

2.8 Acknowledged Gaps and Honest Unknowns

Intellectual honesty requires identifying where our theoretical grounding is strong, where it is extrapolated, and where it faces genuine gaps.

What is well-grounded: The constitutive role of analogy in cognition (Assumption 1) rests on Hofstadter and Sander's extensive empirical and philosophical work [3], supported by converging evidence from Lakoff and Johnson [1] and the broader cognitive science literature on analogical reasoning. The speed advantage of pattern-based retrieval is supported by Slotnick's neuroscience evidence [11].

What is our own contribution: The reframing of analogical cognition as "compressed latent computation" (Assumption 2) is our own contribution and the interpretation to mechanism process this paper undertakes. It is supported by the phenomenology (analogy is fast, automatic, and compressive) yet the computational characterization is ours, not Hofstadter and Sander's. They describe what analogy does in rich cognitive detail; we are the ones proposing that this functional signature can be captured by a learnable geometric architecture.

What remains genuinely open: Whether semantic opposition (SCT's 180° rotation) is best understood as a form of analogical relationship. Hofstadter and Sander's account focuses on similarity-based linking, situations that share "an essence in common" [3], rather than on how opposites relate. The decision to encode opposition as a geometric operation (rotation by π) is an engineering commitment that requires empirical validation, not a claim grounded in the cognitive science literature we cite. Whether analogies can be formally composed, the output of one mapping becoming the input to another, is not addressed by Hofstadter and Sander, though their account of reusable "conceptual skeletons" that can unify "a family of analogies" [3] is suggestive. Whether cultural variation produces systematically different analogical mappings is plausible but unattested in the sources we cite [3].

These gaps are not weaknesses to be papered over; they are **empirical questions that SCT is designed to help answer**. If opposition at 180° produces reliable antonyms across contexts, that is evidence for opposition-as-geometric-relationship. If it does not, we learn something about the limits of angular semantics. The architecture is a hypothesis made concrete.

Finally, one limit is worth stating plainly because it is easy to overread the cognitive-science framing. Mapping cognitive structural alignment to Euclidean angular rotation is an engineered approximation, not a neurobiological claim. Hofstadter and Sander's account of analogy as structural mapping does not entail any specific geometric implementation. Our hypothesis is that this forced geometric regularization yields efficient, steerable representations useful for the navigational tasks SCT is designed to support — not that it models biological cognition.

2.9 Relation to Concurrent Geometric Approaches

SCT belongs to a fast-growing line of work that treats meaning as geometry and relations as rotations, and it is worth situating against that field before turning to the formalism. The closest established precedent is RotatE [5] and its hyperbolic and quaternionic descendants, which model relations as rotations for knowledge-graph link prediction over discrete entities. Concurrently with this paper's development, several groups have proposed rotation as a steering or transformation primitive in continuous representation space: most directly RISE [12], which represents discourse-level semantic-syntactic transformations as post-hoc rotors on the embedding hypersphere, and a cluster of activation-space methods such as Angular Steering [13] and Spherical Steering [14] that rotate hidden activations to control generation. We arrived at SCT's design independently and concurrently — what was a sparse field in mid-2025 (the author's prior-art search at the time surfaced essentially only RotatE) has since become a crowded one — and we make this acknowledgment, and the distinction between approaches, as of June 2026.

SCT is distinguished from all of these on the same axis. It is neither a link-prediction model over static triples nor a post-hoc or activation-level steering operator applied during generation.

It is a non-autoregressive *semantic actuator*: a trained primitive that, given a concept and a parameterized query (angle 0°–360°, magnitude 0–1, context), returns a transformed embedding by projecting into a per-context shadow plane, rotating, and lifting. Its distinctive commitments are the per-context shadow-plane factorization that isolates polysemy, a navigable angular address space — a compass over fixed directions — rather than a single learned transformation, and magnitude as an explicit, continuous control over semantic distance. A detailed comparison against each prior and concurrent system is given in Appendix D.

3. Mathematical Formalization

3.1 Core Transformation Function

Let $S \subset \mathbb{R}^d$ be a d-dimensional semantic embedding space where $d = 768$. We define the transformation function T as:

$\mathbf{T: S \times \Theta \times M \times C \to S}$

where: - $\mathbf{c} \in \mathbf{S}$: input concept embedding - $\boldsymbol{\theta} \in \mathbf{[0, 2\pi]}$: transformation angle (continuous) - $\mathbf{m} \in \mathbf{[0, 1]}$: magnitude (semantic distance) - $\mathbf{x} \in \mathbf{C}$: context from finite set C with $|C| = 12$

3.2 The Shadow Plane Architecture

The architecture employs a factorized projection system that separates **adaptation** from **rotation**. Each context x is equipped with:

Shadow Plane $\mathbf{S_x}$: A learned 2D subspace of $\mathbb{R}^d$, specified by an orthonormal basis, that absorbs embedding-specific geometry.

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Shadow basis: s₁ˣ, s₂ˣ ∈ ℝᵈ where s₁ˣ · s₂ˣ = 0 and ||s₁ˣ|| = ||s₂ˣ|| = 1
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Rotation Plane $\mathbf{P_x}$ *(Shadow-affine configuration only)*: A purely abstract 2D coordinate system in which rotation is expressed. The rotation plane has no embedding into $\mathbb{R}^d$; it is defined only by its relationship to the shadow plane via the alignment transform. The Shadow-direct configuration has no rotation plane: rotation is applied directly in shadow coordinates.

Alignment Transform $\mathbf{W_x}$ *(Shadow-affine configuration only)*: A learned 2×2 mapping between shadow and rotation coordinates.

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Wₓ ∈ ℝ^(2×2) where det(Wₓ) ≠ 0 (invertibility requirement)
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3.3 Forward Transformation Pipeline

The pipeline is presented for both configurations as a single sequence. Steps 2 and 4 apply only to **Shadow-affine** and are skipped entirely under **Shadow-direct**, which rotates directly in shadow coordinates.

Step 1: Project to Shadow Plane

$$\alpha_s = c \cdot s_1^x$$
$$\beta_s = c \cdot s_2^x$$

Step 2: Align to Rotation Plane *(Shadow-affine only)*

$$\begin{bmatrix} \alpha_p \\ \beta_p \end{bmatrix} = W_x \begin{bmatrix} \alpha_s \\ \beta_s \end{bmatrix}$$

Step 3: Rotate

Let (α, β) denote the coordinates being rotated: the shadow coordinates (α_s, β_s) for Shadow-direct, or the rotation-plane coordinates (α_p, β_p) for Shadow-affine.

$$\alpha' = \alpha \cos(\theta) - \beta \sin(\theta)$$
$$\beta' = \alpha \sin(\theta) + \beta \cos(\theta)$$

Step 4: Inverse Align to Shadow Plane *(Shadow-affine only)*

$$\begin{bmatrix} \alpha'_s \\ \beta'_s \end{bmatrix} = W_x^{-1} \begin{bmatrix} \alpha'_p \\ \beta'_p \end{bmatrix}$$

(Under Shadow-direct, the Step 3 outputs (α', β') are already the shadow-coordinate results (α'_s, β'_s).)

Step 5: Compute Delta and Lift

$$\Delta\alpha_s = \alpha'_s - \alpha_s$$
$$\Delta\beta_s = \beta'_s - \beta_s$$
$$c' = c + m \cdot (\Delta\alpha_s \cdot s_1^x + \Delta\beta_s \cdot s_2^x)$$

This formulation ensures: - **Identity at m=0**: No movement regardless of angle - **Identity at θ=0**: No rotation regardless of magnitude - **Approximate involution**: Rotating 180° twice returns approximately to origin; exact return at $m \in \{0, 1\}$; see Section 3.6 Property 2 and Appendix A.1 for the precise statement. - **Magnitude monotonicity**: Larger m ⟶ larger semantic shift - **Context isolation**: Training on context x leaves context y's context-specific parameters untouched

3.4 Alignment Configurations

SCT admits two architectural configurations that differ in whether an alignment transform sits between projection and rotation.

3.4.1 Shadow-direct

Rotation is applied directly in shadow-plane coordinates; there is no alignment transform and no rotation plane. The shadow basis (s_1^x, s_2^x) carries all per-context orientation responsibility, and this configuration adds **zero** alignment parameters per context.

```
Rotate (αₛ, βₛ) by θ; lift the magnitude-scaled delta back through the shadow basis.
```

Invertibility: Not applicable — there is no alignment matrix to invert. The rotation $R(\theta) \in SO(2)$ is its own well-defined inverse $R(-\theta)$.

3.4.2 Shadow-affine

A learned alignment transform $W_x \in GL(2)$ maps shadow coordinates into a rotation plane; rotation is applied there, and the inverse transform returns to shadow coordinates before the lift. This adds 4 parameters per context (48 total) and admits rotation, scaling, and shear — for contexts whose embedding geometry deviates from rotation-friendly structure.

Invertibility Enforcement: We parameterize W_x via its SVD to guarantee $\det(W_x) > 0$:

```
# W = U @ diag(σ) @ Vᵀ where σ > 0
σ = softplus(σ_raw) + ε  # Ensures positive singular values
W = U @ diag(σ) @ V.T    # Always invertible
```

3.4.3 Why no orthogonal (SO(2)) alignment variant

An intermediate option — constraining W_x to a pure rotation in SO(2) — is deliberately **excluded**, because it would be mathematically inert. SO(2) is abelian, so for any $W_x \in SO(2)$ the alignment conjugates away:

```
Wₓ⁻¹ R(θ) Wₓ = R(θ)
```

The alignment angle would therefore have no effect on the transform's output and would constitute dead parameters. The simpler Shadow-direct path (no alignment at all) is exactly equivalent to it, so Shadow-direct is the configuration we keep. Only a non-orthogonal alignment — Shadow-affine, with scale and shear — changes the operation. This is why the two retained configurations are Shadow-direct and Shadow-affine, with no orthogonal variant between them.

> **Implementation Note**: The choice between configurations is a hyperparameter to be determined empirically. Shadow-direct is the default for its parameter economy and interpretability; Shadow-affine is the upgrade path if per-context accuracy metrics indicate that scale/shear expressivity is needed. See Section 6 for ablation methodology.

3.5 Proximity-Gated Attention Mechanism

Each head $h \in \{1,\ldots,8\}$ is assigned a compass direction θ_h: - Heads 1-4: Cardinal directions {0°, 90°, 180°, 270°} - Heads 5-8: Intercardinal directions {45°, 135°, 225°, 315°}

The head's attention score is modulated by angular proximity, computed on the circle (so that the 0°/360° seam is handled correctly):

Angular Distance (Modular):

```
d(θ, θₕ) = min(|θ - θₕ|, 2π - |θ - θₕ|)
```

This is the shorter arc between θ and θ_h on the unit circle, bounded in $[0, \pi]$.

Angular Proximity Function:

```
pₕ(θ) = max(0, 1 - d(θ, θₕ)/(π/4))
```

Equivalent torch implementation:

```
d = torch.abs(torch.remainder(θ - θₕ + math.pi, 2*math.pi) - math.pi)
p_h = torch.clamp(1 - d / (math.pi/4), min=0)
```

This formulation restores the "sum to 1 between adjacent heads" property everywhere on the circle, including across the 0°/360° boundary.

Proximity-Gated Q/K/V Projections:

```
Qₕ = pₕ(θ) · (x W_q^h)  [48D]
Kₕ = pₕ(θ) · (x W_k^h)  [48D]
Vₕ = pₕ(θ) · (x W_v^h)  [96D]
```

V-Signal Interpolation:

```
V_out = Σₕ softmax(QₕKₕᵀ/√d_k) Vₕ
```

This ensures heads co-activate continuously for intermediate angles θ within π/4 of any head direction, with no dead zones at the 0°/360° seam.

3.6 Geometric Properties and Invariants

Property 1 (Identity at Zero):

```
T(c, 0, m, x) = c  ∀m, x  // No rotation
T(c, θ, 0, x) = c  ∀θ, x  // No magnitude
```

Property 2 (Involution):

The shadow plane architecture geometrically guarantees involution at $m \in \{0, 1\}$, with bounded residual $4m(1-m) \cdot \|proj\_shadow(c)\|$ at intermediate magnitudes. This is an architectural inductive bias, not a learned property: training learns which embeddings to project, not whether the rotation reverses. Concretely, exact involution holds at the endpoints of the magnitude interval:

```
T(T(c, π, 0, x), π, 0, x) = c   // m = 0, trivially
T(T(c, π, 1, x), π, 1, x) = c   // m = 1
```

For intermediate magnitudes $m \in (0, 1)$, involution is approximate with residual:

```
||T(T(c, π, m, x), π, m, x) - c|| = 4m(1-m) · ||proj_shadow(c)||
```

where `proj_shadow(c) = (c · s₁ˣ) · s₁ˣ + (c · s₂ˣ) · s₂ˣ` is the component of c lying within the shadow plane S_x. The residual is maximized at $m = 0.5$, where it equals $\|proj\_shadow(c)\|$ exactly, and vanishes when c is orthogonal to the shadow plane. See Appendix A.1 for derivation.

The exact-endpoint property rests on: - $R(\pi) \cdot R(\pi) = I$ in the rotation plane - $W_x^{-1} \cdot W_x = I$ ensures alignment cancels - $m \in \{0, 1\}$ ensures full cancellation at the lift step

In practice, opposition ($\theta = \pi$) is evaluated at m = 1 in production, where involution is exact. Intermediate m values correspond to interpolated semantic shifts, where small residuals are acceptable and empirically quantified in Section 6.1.

Property 3 (Context Isolation):

Context-specific parameters, the shadow basis vectors s_1^x, s_2^x and alignment matrix W_x, receive zero gradient from examples in other contexts. Context y's s_1^y, s_2^y, and W_y are unaffected by training on context x.

Shared parameters, the embedding table and the compass attention stack (attention projections, feedforward layers, layer norms), receive gradients from all contexts. The embedding table's 1e-6 learning rate heavily attenuates this cross-context pressure; the transformer components (attention and FFN) adapt at standard learning rates and are expected to develop context-general representations through exposure to all contexts. Context-specificity in rotation semantics is therefore concentrated entirely in the shadow adapter, by design.

Property 4 (Angular Semantics): - **θ = 0°**: Identity/Reinforcement - **θ = 90°**: Analogy/Orthogonal concept - **θ = 180°**: Opposition/Antonym - **θ = 270°**: Complement/Inversion

Intercardinal angles (45°, 135°, 225°, 315°) interpolate between these cardinal semantics.

4. Architecture Specification

4.1 Complete Architecture (~131M Parameters)

```
Input Processing:
├── Text Tokenization (max_length=32)
├── Embedding Layer (vocab_size=120k, dim=768)
├── Positional Encoding (max_length=32, dim=768)
├── Geometric Encoding
│   ├── Angle Encoding: [cos(θ), sin(θ)] → Linear(2, 768)
│   ├── Magnitude Encoding: m → Linear(1, 768)
│   └── Context Encoding: one-hot(x) → Embedding(12, 768)
└── Input Fusion: LayerNorm(c + θ_enc + m_enc + x_enc)\

Shadow Plane Adapter:
├── Shadow Basis Vectors: {s₁ˣ, s₂ˣ} for each x ∈ 12 contexts
└── Alignment Transform Wₓ (Shadow-affine configuration only)
    ├── Shadow-affine: SVD-parameterized GL(2) (4 params/context)
    └── Shadow-direct: no alignment (rotation in shadow coordinates)

Compass Attention Core (6 Layers):
├── CompassHeads (×8 per layer)
│   ├── Heads 1-4: Cardinal directions (0°, 90°, 180°, 270°)
│   ├── Heads 5-8: Intercardinal directions (45°, 135°, 225°, 315°)
│   └── Uneven Q/K/V Split: Q/K=48D, V=96D per head
├── Proximity Gating (modular angular distance)
├── LayerNorm & Residual
└── FeedForward (768 → 3072 → 768)

Geometric Transform Module:
├── Shadow Projection: c → (αₛ, βₛ)
├── Alignment (Shadow-affine only): (αₛ, βₛ) → (αₚ, βₚ) via Wₓ
├── Rotation: R(θ) — in shadow coordinates (Shadow-direct) or rotation plane
(Shadow-affine)
├── Inverse Alignment (Shadow-affine only): (α'ₚ, β'ₚ) → (α'ₛ, β'ₛ) via Wₓ⁻¹
└── Magnitude-Scaled Lift: c' = c + m·Δ
```

```
Output Head:
├── TransformedEmbedding → ℝ^768
├── Temperature-controlled selection (τ ∈ [0.01, 0.5])
└── Nearest Neighbor → discrete token from 120k vocab
```

4.2 Parameter Budget

Component	Parameters	Notes
Embedding Layer	92.16M	120k × 768 (full lexical discrimination)
FeedForward (6 layers)	28.31M	6 × (768×3072 + 3072×768)
Compass Heads (6 layers × 8 heads)	10.62M	6 × (W_q + W_k + W_v + W_o); uneven 48/48/96 split
LayerNorm / Biases	~57K	13 LN instances + attn+FFN biases across 6 layers
Positional Encoding	~25K	32 × 768 learned
Shadow Basis Vectors	18.4K	2 × 768 × 12 contexts
Geometric Encoders	~13K	Angle: 2⟶768; magnitude: 1⟶768; context embedding: 12×768
Alignment Matrices	0 (Shadow-direct) / 48 (Shadow-affine)	Shadow-direct: none; Shadow-affine: 4 per context
TOTAL	**~131.2M**	**70% embedding, 22% FFN, 8% attention, <0.1% adapter+encoding**

The parameter distribution reflects SCT's architectural philosophy: the 120k × 768 embedding table carries the lexical discrimination load, the transformer stack handles contextual processing, and the shadow adapter performs the per-context rotation-anchoring role with minimal parameter footprint. The adapter's small size is a feature of its narrow role, not a symptom of underprovisioning; context-specific expressive capacity beyond rotation-anchoring is carried by other components (attention, FFN, and the context-conditioned input fusion).

4.3 Learning Rate Hierarchy

The shadow plane architecture creates a natural hierarchy of adaptation rates. Components closer to the "core" (embeddings) should move slowly to preserve pre-trained structure; components at the "periphery" (alignment matrices) can adapt quickly.

```
learning_rates = {
    # Core: preserve pre-trained structure
    "embeddings": 1e-6,

    # Medium: primary context-specific adaptation surface
    "shadow_basis": 5e-4,

    # Fast: fine-grained per-context adjustment
    "alignment": 1e-3,

    # Standard: transformer components
    "attention": 3e-4,
    "feedforward": 3e-4,
}
```

> **Tuning Note**: These rates represent starting points derived from theoretical considerations. Production deployment may require adjustment based on convergence behavior and validation metrics. The key invariant is the **ratio** between rates: embeddings should move ~500× slower than shadow projections; alignment should move ~2× faster than shadow projections. Absolute values may scale together.[1]

4.4 Loss Function

```
L_total = λ_transform · L_transform
        + λ_geometric · L_geometric
        + λ_proximity · L_proximity
        + λ_continuity · L_continuity
        + λ_specialization · L_specialization
        + λ_alignment · L_alignment
        + λ_orthogonality · L_orthogonality
```

Of these terms, **L_alignment** is specific to the **Shadow-affine** configuration — it regularizes the alignment transform W_x, which exists only there — and is omitted under **Shadow-direct**. All other terms apply to both configurations.

L_transform: Cross-entropy on target token (primary objective)

L_geometric: Identity and involution constraints, enforced only where they hold exactly:

```
L_geometric = ||T(c, 0, m, x) - c||²                        (identity at θ=0,
all m)
            + ||T(T(c, π, 1, x), π, 1, x) - c||²         (involution at m=1)
            + monotonicity_penalty
```

The involution term is evaluated only at m = 1, where exact involution holds (Property 2). Evaluating it at arbitrary m ∈ (0, 1) would train the model against its own geometry: the involution residual at intermediate m is a structural 4m(1-m)·‖proj_shadow(c)‖ that cannot be driven to zero without collapsing proj_shadow(c) ⟶ 0, which would eliminate rotation entirely. Restricting enforcement to m = 1 avoids this pathology.

L_proximity: Reward proportional to angular closeness

```
L_proximity = Σₕ ||headₕ_attention - p_h(θ)||²
```

where p_h(θ) uses the modular angular distance from Section 3.5.

L_continuity: V-signal smoothness

```
L_continuity = Σ||∂V/∂θ||²
```

L_specialization: Head entropy maximization

```
L_specialization = -Σ head_attention · log(head_attention + ε)
```

L_alignment: Alignment matrix regularization (Shadow-affine only):

```
L_alignment = Σₓ ||Wₓ - I||²_F
```

L_orthogonality: Shadow basis orthonormality:

```
L_orthogonality = Σₓ (|s₁ˣ · s₂ˣ|² + (||s₁ˣ|| - 1)² + (||s₂ˣ|| - 1)²)
```

Recommended λ values:

Term	λ	Rationale
transform	1.0	Primary objective
geometric	0.5	Hard constraints
proximity	0.3	Compass training
continuity	0.1	Smoothness
specialization	0.1	Head diversity
alignment	0.05	Minimal distortion (Shadow-affine)
orthogonality	1.0	Hard constraint

5. Training Methodology

5.1 Training Data Generation

Total corpus: **~200,000 unique transformations** (~50M tokens)


```json
{
  "input": {
    "concept": "hot",
    "transform_angle": 37,
    "transform_magnitude": 0.9,
    "transform_context": "physics"
  },
  "target": "warm",
  "cardinal_proximities": {
    "0": 0.18,
    "45": 0.82,
    "90": 0.00,
    "135": 0.00,
    "180": 0.00,
    "225": 0.00,
    "270": 0.00,
    "315": 0.00
  },
  "generation_metadata": {
    "consensus_models": ["claude-opus-4-1", "sonnet-4-5", "deepseek-v3"],
    "agreement_score": 0.76
  }
}
```


Multi-LLM consensus pipeline with cost optimization (~$1,500 budget): - High-Precision (Medical/Legal): Claude Opus 4.1 - Standard Precision (Physics/Business): Sonnet 4.5 - Bulk/Cultural: Gemini Flash / DeepSeek V3 / Qwen

5.2 Training Configuration

```
config = {
    "batch_size": 128,
    "learning_rate": {
        "embeddings": 1e-6,
        "shadow_basis": 5e-4,
        "alignment": 1e-3,
        "transformer": 3e-4,
    },
    "shadow_plane": {
        "alignment_variant": "direct",  # or "affine"
        "alignment_lambda": 0.05,
        "orthogonality_lambda": 1.0,
    },
    "mixed_precision": "fp16",
    "gradient_accumulation": 2,
    "max_epochs": 75,
    "training_time_estimate": "18-22 hours on RTX 3090"
}
```

5.3 Lexical Distribution Strategy

We reject artificial summation to 100%. Words are polysemous semantic vectors; their participation in multiple contexts is a feature of geometric navigation:

Context	Core Terms	Extended	% of 120k
philosophy	4,000	12,000	10%
judiciary	6,000	15,000	12.5%
physics	5,000	10,000	8.3%
mathematics	3,500	8,000	6.7%
symbolic-logic	800	2,500	2.1%
psychology	5,500	12,000	10%
general-medicine	12,000	25,000	20.8%
business-culture	3,500	10,000	8.3%
popular-culture	15,000	35,000	29.2%
urban-culture	8,000	20,000	16.7%
rural-culture	4,500	12,000	10%
romantic-relations	2,500	6,000	5%
TOTAL	**70,300**	**167,500**	**139.6%**

The shadow plane architecture makes this overlap tractable: each context's shadow projection reframes the same embedding point for its specific rotation semantics, without pushing the point itself around.

6. Evaluation Methodology

6.1 Geometric Sanity Checks

Involution Accuracy at m = 1 (exact-return regime):

```
involution_accuracy = mean(cosine_sim(c, T(T(c, π, 1, x), π, 1, x)))
Target: >0.98
```

Involution Residual at m ∈ (0, 1) (bounded-return regime):

Verify empirically that residuals track the theoretical bound:

```
for m in [0.25, 0.5, 0.75]:
    empirical_residual = mean(||T(T(c, π, m, x), π, m, x) - c||)
    theoretical_bound = 4*m*(1-m) * mean(||proj_shadow(c)||)
Expect: empirical_residual ≤ theoretical_bound with small slack
```

Magnitude Monotonicity:

```
for m in [0, 0.2, 0.4, 0.6, 0.8, 1.0]:
    d_m = ||T(c, θ, m, x) - c||
Expect: Monotonically increasing
```

Shadow Basis Orthonormality:

```
shadow_ortho = mean(|s₁ˣ · s₂ˣ|) across all x
shadow_norms = mean(|||s₁ˣ|| - 1| + |||s₂ˣ|| - 1|) across all x
Target: both < 0.01
```

Angular Resolution:

```
Count distinct outputs for θ ∈ {0°, 5°, 10°, ..., 355°}
Target: >24 loci at τ=0.1; >48 loci at τ=0.05; >64 loci at τ=0.01
```

6.2 Semantic Quality Metrics

Metric	Target	Measurement
Opposition Accuracy (180°)	>96%	Correct antonyms
Analogy Accuracy (90°)	>94%	LLM-validated orthogonal relations
Intercardinal Accuracy	>92%	Correct at 45°, 135°, 225°, 315°
Context Consistency	Qualitative	Same (θ,m) yields related concepts across contexts
Interpolation Smoothness	Qualitative	No discontinuous jumps in magnitude sweep

6.3 Ablations

The ablation ladder isolates each inductive bias by removing it and measuring the effect. Each configuration uses the same frozen embedding model, the same 120k vocabulary, the same nearest-neighbor decoding, and the same training data.

1. **MLP baseline**: No geometric prior at all. `[concept_embedding ⊕ sin(θ) ⊕ cos(θ) ⊕ magnitude ⊕ one-hot(context)] → 3-layer MLP → predicted_embedding`. Tests whether the geometric inductive bias provides any advantage over unconstrained learning — the direct answer to whether the architecture buys anything beyond a sufficiently parameterized function approximator.
2. **No shadow plane**: Rotate directly in the full 768D embedding space. Tests whether the shadow plane is buying anything over rotation in the native space.
3. **Shadow-direct**: Shadow plane plus rotation, no alignment matrix. Tests whether the shadow plane alone is sufficient.
4. **Shadow-affine**: Shadow plane plus SVD-parameterized GL(2) alignment plus rotation. Tests whether non-rotational alignment expressivity (scale, shear) adds value beyond the shadow plane.

Compass head count (8 vs. 4 vs. no compass attention) is a further planned ablation, isolating the contribution of the proximity-gated attention mechanism.

Metrics: - Per-context accuracy breakdown - Cross-context interference (train on x, measure degradation on y) - Convergence speed - Plane separation matrix $\Omega(S_x, S_y)$

Comparisons examine raw accuracy, sample efficiency curves across training-set sizes, and generalization to held-out angular ranges. We do not commit to specific dataset sizes or angular ranges in this position paper; those are fixed in the companion empirical paper.

6.4 Plane Separation Analysis

After training, compute:

```
def plane_separation_matrix(shadow_adapter):
    Omega = torch.zeros(12, 12)
    for x in range(12):
        for y in range(12):
            s1x, s2x = shadow_adapter.shadow_basis[x]
            s1y, s2y = shadow_adapter.shadow_basis[y]
            Omega[x,y] = (abs(s1x@s1y) + abs(s1x@s2y) +
                          abs(s2x@s1y) + abs(s2x@s2y))
    return Omega
```

Analyze: - Do related contexts (physics-chemistry) show high overlap? - Do unrelated contexts (physics-mythology) show low overlap? - Does overlap correlate with per-context accuracy?

6.5 Falsifiability Criteria

The SCT hypothesis should be considered falsified if (a) MLP baselines match SCT on held-out angle generalization, (b) sample efficiency curves are indistinguishable across architectures, (c) per-context accuracy fails to correlate with shadow plane separation, or (d) involution residuals at intermediate magnitudes systematically exceed the 4m(1−m)·‖proj_shadow(c)‖ bound after training. We commit to reporting these comparisons whether or not they favor the proposed architecture.

7. Implementation

7.1 Shadow Plane Adapter Module

```
class ShadowPlaneAdapter(nn.Module):
    def __init__(
        self,
        embed_dim: int = 768,
        num_contexts: int = 12,
        alignment_variant: str = "direct",
    ):
        super().__init__()
        self.embed_dim = embed_dim
        self.num_contexts = num_contexts
        self.alignment_variant = alignment_variant

        # Shadow basis: always learned
        self.shadow_basis = nn.Parameter(
            torch.randn(num_contexts, 2, embed_dim)
        )

        # Alignment matrices exist only in the Shadow-affine configuration.
          # Shadow-direct uses no alignment transform and adds no alignment
parameters.
        if alignment_variant == "affine":
            # SVD-parameterized (Section 3.4.2) for guaranteed invertibility
            self.alignment_matrices = nn.Parameter(
                torch.eye(2).unsqueeze(0).repeat(num_contexts, 1, 1)
            )

        self._initialize_orthonormal()

    def _initialize_orthonormal(self):
        with torch.no_grad():
            for x in range(self.num_contexts):
```

```
                v1 = self.shadow_basis[x, 0]
                v1 = v1 / torch.norm(v1)
                v2 = self.shadow_basis[x, 1]
                v2 = v2 - (v2 @ v1) * v1
                v2 = v2 / torch.norm(v2)
                self.shadow_basis[x, 0] = v1
                self.shadow_basis[x, 1] = v2

    def get_alignment(self, context_idx):
        # Defined only for Shadow-affine; Shadow-direct never calls this.
        W = self.alignment_matrices[context_idx]
        W_inv = torch.inverse(W)
        return W, W_inv

    def forward(self, embeddings, angle, magnitude, context_idx):
        # Step 1: Project to shadow plane
        s1, s2 = self.shadow_basis[context_idx]
        alpha_s = embeddings @ s1
        beta_s = embeddings @ s2
        shadow_coords = torch.stack([alpha_s, beta_s], dim=-1)

        cos_a = torch.cos(angle).unsqueeze(-1)
        sin_a = torch.sin(angle).unsqueeze(-1)

        if self.alignment_variant == "affine":
            # Step 2: Align to rotation plane
            W, W_inv = self.get_alignment(context_idx)
            rotation_coords = shadow_coords @ W.T
            # Step 3: Rotate in rotation plane
          alpha_r, beta_r = rotation_coords[..., 0:1], rotation_coords[..., 1:2]
            alpha_r_rot = alpha_r * cos_a - beta_r * sin_a
            beta_r_rot = alpha_r * sin_a + beta_r * cos_a
            rotation_rot = torch.cat([alpha_r_rot, beta_r_rot], dim=-1)
            # Step 4: Inverse align back to shadow plane
```

```
            shadow_rot = rotation_rot @ W_inv.T
        else:
         # Shadow-direct: Steps 2 and 4 are skipped; rotate in shadow coordinates
            alpha_d, beta_d = shadow_coords[..., 0:1], shadow_coords[..., 1:2]
            alpha_d_rot = alpha_d * cos_a - beta_d * sin_a
            beta_d_rot = alpha_d * sin_a + beta_d * cos_a
            shadow_rot = torch.cat([alpha_d_rot, beta_d_rot], dim=-1)

        # Step 5: Compute delta and lift
        delta = shadow_rot - shadow_coords
        delta_alpha, delta_beta = delta[..., 0:1], delta[..., 1:2]
        lift = delta_alpha * s1 + delta_beta * s2

        # Apply magnitude
      magnitude = magnitude.unsqueeze(-1) if magnitude.dim() == 1 else magnitude
        transformed = embeddings + magnitude * lift

        return transformed
```

7.2 Training Loop Integration

```
def train_step(model, batch, optimizer, config):
    embeddings = model.embed(batch['concept'])
    angle = batch['angle'] * (math.pi / 180)  # Convert to radians
    magnitude = batch['magnitude']
    context = batch['context']

    # Forward through shadow adapter
    transformed = model.shadow_adapter(
        embeddings, angle, magnitude, context
    )

    # Compute losses
    loss_transform = cross_entropy(
        model.decode(transformed),
        batch['target']
    )

    loss_geometric = geometric_constraints(
        model, embeddings, angle, magnitude, context
    )

    loss_proximity = proximity_reward(
        model.compass_attention.head_activations,
        angle,
        model.compass_attention.head_angles
    )

    loss_shadow = shadow_regularization(
        model.shadow_adapter,
        config['shadow_plane']
    )

    loss_total = (
```

```
        config['lambda']['transform'] * loss_transform +
        config['lambda']['geometric'] * loss_geometric +
        config['lambda']['proximity'] * loss_proximity +
        config['lambda']['shadow'] * loss_shadow
    )

    loss_total.backward()
    optimizer.step()

    return {
        'loss_total': loss_total.item(),
        'loss_transform': loss_transform.item(),
        'loss_geometric': loss_geometric.item(),
        'loss_proximity': loss_proximity.item(),
        'loss_shadow': loss_shadow.item(),
    }
```

8. Conclusion

Strange Compass Transformers demonstrate that semantic navigation can be achieved through geometric operations in learned 2D planes with continuous angular resolution. Shadow Plane Adapters solve the polysemy problem by factorizing adaptation from rotation semantics. The philosophical foundations ground the entire enterprise in a stated, testable thesis: that analogical cognition is compressed latent computation, and that SCT can externalize this compression as navigable geometry.

The core insights are:

1. **Analogical structure is constitutive of cognition**: Hofstadter and Sander's work establishes that analogy is not a tool cognition occasionally deploys but the process by which cognition operates, continuous, automatic, and compressive.
2. **Compression produces reusable relational skeletons**: The "conceptual skeleton" mechanism documented by Hofstadter and Sander describes exactly the kind of structure SCT aims to encode: abstract relational patterns that transfer across domains while discarding surface detail.
3. **Compressed latent computation is the paper's interpretive bridge**: We characterize the cognitive process as compressed latent computation, an interpretive move that is ours, grounded in the phenomenology but not claimed as established cognitive science. This framing is what licenses the engineering commitment: if the functional signature of analogical cognition is systematic enough, it can be approximated by a learnable architecture.
4. **Polysemy is solvable via factorization**: Shadow planes absorb context-specific adaptation without corrupting shared embeddings.
5. **Precision is tunable**: Learning rate hierarchies provide graduated control over what adapts and how quickly.
6. **Validation is empirical**: The thesis is concrete enough to be tested. If 180° reliably produces antonyms, if 90° reliably produces analogical relations, if magnitude monotonically controls semantic distance, then the hypothesis that analogical structure can be externalized as navigable geometry gains support. If it does not, we learn something about the limits of the approach. Either outcome advances understanding. The acknowledged limits of the geometric-rotation-as-analogy commitment — that the mapping is an engineered approximation rather than a neurobiological claim — are stated in full in Section 2.8, and SCT is positioned relative to prior and concurrent geometric approaches in Section 2.9 and Appendix D.

A final point on the status of what we have specified. The architecture above is our leading conception of how to realize this primitive, not a fixed commitment to its particular mechanisms. Should construction and training reveal that some component fails to do the work asked of it, the design will change; the constant across any such revision is the thesis under test — that analogical structure can be externalized as navigable geometry — and the goal of building a primitive that serves it. We follow the data toward whatever architecture best realizes that goal.

This approach enables systematic exploration of meaning space with 64+ angular resolution points, providing a new primitive for AI reasoning systems that complements rather than replaces existing language models.

Footnotes

[1] **On Learning Rate Tuning**: The recommended rates (embeddings: 1e-6, shadow: 5e-4, alignment: 1e-3) derive from the principle that adaptation should flow from periphery to core. Absolute values may require adjustment based on batch size, precision mode, and optimizer choice. The invariant to preserve is the ratio structure: alignment moves fastest, shadow moves moderately, embeddings barely move. If convergence is unstable, scale all rates down proportionally. If convergence is too slow, scale up.

[2] **On Hofstadter's Spatial Language**: The observation that Hofstadter constructs his argument about analogy through pervasively spatialized metaphors emerged through the author's close reading of *Surfaces and Essences* and subsequent dialogue across multiple AI systems during the development of SCT's theoretical framing. See Acknowledgments for further context.

Acknowledgments

The Strange Compass Transformer is a product of the "Internet of models," strengthened by multiple perspectives shaping its development.

The philosophical framing in Section 2 crystallized through sustained engagement with Hofstadter and Sander's *Surfaces and Essences* [3], supported by structured evidence extraction and synthesis. The observation that Hofstadter's own argument about analogy relies pervasively on spatial metaphors (centers and peripheries, neighborhoods of meaning, distances between concepts) emerged through close reading and informed the conviction that relational structure between concepts has navigable geometric properties. We thank Hofstadter and Sander for providing the cognitive science foundation upon which SCT's philosophical warrant rests, while noting that SCT's specific architectural commitments (angular semantics, shadow plane adapters, compass attention) are original engineering decisions not derived from or endorsed by their work.

We also thank the open-source embedding model community, particularly Nomic AI for their fully reproducible training methodology, which informed our approach to transparency and replicability.

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Appendix A: Detailed Geometric Derivations

A.1 Involution Under the Shadow Plane Architecture

We establish here that the shadow plane architecture geometrically enforces involution at $m \in \{0, 1\}$, with predictable residual at intermediate magnitudes. This is a structural property of the layer, not an empirical claim about learned behavior.

Derivation:

The derivation below is written for the Shadow-affine configuration, with the alignment transform W_x explicit. It specializes to Shadow-direct by setting $W_x = I$ (Steps 2 and 4 become identities); the result is unchanged.

Let `proj_shadow(c) = αₛ · s₁ˣ + βₛ · s₂ˣ`, where $\alpha_s = c \cdot s_1^x$ and $\beta_s = c \cdot s_2^x$. This is the component of c lying within the shadow plane S_x. By the orthonormality of $\{s_1^x, s_2^x\}$, we have $\|\text{proj\_shadow}(c)\|^2 = \alpha_s^2 + \beta_s^2$.

First application of $T(\cdot, \pi, m, x)$ on c:

```
Step 1: Project
  αₛ = c · s₁ˣ, βₛ = c · s₂ˣ

Step 2: Align to rotation plane
  [αₚ, βₚ]ᵀ = Wₓ [αₛ, βₛ]ᵀ

Step 3: Rotate by π
  α'ₚ = -αₚ, β'ₚ = -βₚ    (cos π = -1, sin π = 0)

Step 4: Inverse align
  [α'ₛ, β'ₛ]ᵀ = Wₓ⁻¹ [-αₚ, -βₚ]ᵀ = -Wₓ⁻¹ Wₓ [αₛ, βₛ]ᵀ = -[αₛ, βₛ]ᵀ

Step 5: Delta and lift
  Δαₛ = α'ₛ - αₛ = -2αₛ,  Δβₛ = -2βₛ
  c₁ = c + m · (Δαₛ · s₁ˣ + Δβₛ · s₂ˣ)
     = c - 2m · (αₛ · s₁ˣ + βₛ · s₂ˣ)
     = c - 2m · proj_shadow(c)
```

Second application of $T(\cdot, \pi, m, x)$ on c_1:

The shadow-plane projection of c_1 is:

```
α₁ = c₁ · s₁ˣ = αₛ - 2m · αₛ = αₛ(1 - 2m)
β₁ = c₁ · s₂ˣ = βₛ(1 - 2m)
```

(using $\|s_1^x\| = \|s_2^x\| = 1$ and $s_1^x \cdot s_2^x = 0$)

Following the same steps 2–4 on (α_1, β_1), the deltas are:

```
Δα = -2α₁ = -2αₛ(1 - 2m)
Δβ = -2β₁ = -2βₛ(1 - 2m)
```

The second lift yields:

```
c₂ = c₁ + m · (Δα · s₁ˣ + Δβ · s₂ˣ)
   = c₁ - 2m(1 - 2m) · (αₛ · s₁ˣ + βₛ · s₂ˣ)
   = c₁ - 2m(1 - 2m) · proj_shadow(c)
```

Combining with the c_1 expression:

```
c₂ = c - 2m · proj_shadow(c) - 2m(1 - 2m) · proj_shadow(c)
   = c - 2m · [1 + (1 - 2m)] · proj_shadow(c)
   = c - 2m · (2 - 2m) · proj_shadow(c)
   = c - 4m(1 - m) · proj_shadow(c)
```

Cases: - $m = 0$: $c_2 = c$ (trivially) - $m = 1$: $c_2 = c - 4(1)(0) \cdot \text{proj\_shadow}(c) = c$ ▮ - $m \in (0, 1)$: $c_2 = c - 4m(1 - m) \cdot \text{proj\_shadow}(c)$; exact residual is $4m(1-m) \cdot \|\text{proj\_shadow}(c)\|$

Residual bound: For all $m \in [0, 1]$,

```
||c₂ - c|| = 4m(1-m) · ||proj_shadow(c)|| ≤ ||proj_shadow(c)||
```

with equality at $m = 0.5$. Residual vanishes as $\|\text{proj\_shadow}(c)\| \to 0$, i.e., as c becomes orthogonal to the shadow plane.

Practical interpretation: Opposition ($\theta = \pi$) at $m = 1$ is exact by construction. For intermediate magnitudes, the architecture is self-consistent only up to a predictable, bounded residual; this residual should not be driven to zero by training, since doing so would require proj_shadow(c) $\to 0$, eliminating the shadow plane's role entirely. Section 4.4's L_geometric therefore enforces involution only at $m = 1$, where it is mathematically exact.

A.2 Alignment Invertibility

Alignment invertibility is required only in the **Shadow-affine** configuration; **Shadow-direct** has no alignment matrix and nothing to invert (its rotation $R(\theta) \in SO(2)$ is its own inverse $R(-\theta)$).

Shadow-affine. Under the SVD parameterization specified in Section 3.4.2:

```
W = U · diag(σ) · Vᵀ
```

where $U, V \in SO(2)$ and $\sigma = \text{softplus}(\sigma\_raw) + \varepsilon$ with $\varepsilon > 0$ ensures both singular values satisfy $\sigma_i > 0$. Therefore:

```
det(W) = det(U) · σ₁·σ₂ · det(Vᵀ) = 1 · σ₁σ₂ · 1 = σ₁σ₂ > 0  ∎
```

The matrix is always invertible, with $W^{-1} = V \cdot \text{diag}(1/\sigma) \cdot U^T$ computable in closed form without numerical inversion. The Shadow-affine alignment therefore guarantees invertibility by construction while admitting rotation, scale, and shear (4 parameters per context).

Appendix B: Hyperparameter Sensitivity Analysis

The following parameters warrant empirical tuning:

Parameter	Default	Range to Explore	Sensitivity
lr_embedding	1e-6	[1e-7, 1e-5]	High (affects drift)
lr_shadow	5e-4	[1e-4, 1e-3]	Medium
lr_alignment	1e-3	[5e-4, 5e-3]	Low
λ_alignment	0.05	[0.01, 0.2]	Low
alignment_variant	direct	{direct, affine}	Context-dependent

Priority order for tuning: (1) lr_embedding, (2) alignment_variant, (3) remaining parameters.

Appendix C: Cognitive Science Evidence Map

The following table maps Section 2 claims to their evidentiary sources, distinguishing between claims grounded in cited literature and claims that represent this paper's original contributions.

Claim	Source	Status
Analogy is constitutive of cognition	Hofstadter & Sander [3]	Directly supported
Analogy = categorization	Hofstadter & Sander [3]	Directly supported
Analogies occur continuously, below awareness	Hofstadter & Sander [3]	Directly supported
Compression via "conceptual skeleton"	Hofstadter & Sander [3]	Directly supported
Scientific discovery driven by analogy	Hofstadter & Sander [3]	Directly supported (9 examples)
Naïve analogies are lossy but functional	Hofstadter & Sander [3]	Directly supported
Pattern-based retrieval faster than deliberative	Slotnick [11]	Supported (ERP timing)
Machines need analogical capacity	Hofstadter & Sander [3]	Directly supported
Conceptual metaphors as cognitive infrastructure	Lakoff & Johnson [1]	Directly supported
"Compressed latent computation" framing	This paper	Original contribution
180° = opposition as analogical relationship	This paper	Original; requires empirical test
Analogies are formally composable	This paper	Original; suggestive precedent in [3]
Cultural variation in analogical mappings	This paper	Plausible; unattested in cited sources

Appendix D: Prior and Concurrent Work — A Detailed Analysis

This appendix expands Section 2.9. It situates SCT against the relevant prior and concurrent literature, distinguishing genuine precedents that require positioning from adjacent work that shares vocabulary but not mechanism. The organizing claim is that SCT occupies a position — a non-autoregressive, per-context, angularly-addressable semantic actuator — that none of the following occupy, even as several share its premise that relations can be expressed as rotations.

D.1 Rotation-based relational embeddings (RotatE and descendants)

RotatE [5] models each relation as an element-wise rotation in complex space, with relation vectors of unit modulus, capturing symmetry, antisymmetry, inversion, and composition through phase angles. Hyperbolic and quaternionic descendants extend the same idea to negatively curved spaces for hierarchical graphs. These are the deepest precedent for "relation as rotation," and SCT shares the intuition that 180° encodes opposition and that relational structure has angular form. The distinction is one of object and task: RotatE and its descendants are link-prediction models over discrete entity/relation triples in a fixed geometric space. SCT operates on continuous embeddings of arbitrary vocabulary, learns a per-context 2D projection rather than a single global space, and is queried as an actuator (concept + angle + magnitude + context ⟶ transformed embedding) rather than scored for triple plausibility. Magnitude as a continuous transformation control has no analogue in the unit-modulus rotation of RotatE.

D.2 Concurrent rotation-as-transformation in embedding space (RISE)

The closest concurrent work is RISE [12], submitted October 2025, which represents discourse-level semantic-syntactic transformations (negation, conditionality, politeness) as consistent rotational operations on the embedding hypersphere and demonstrates cross-lingual, cross-model transfer. RISE and SCT independently reach the conviction that semantic transformations correspond to consistent geometric rotations in embedding space. They differ structurally on nearly every other axis. RISE is post-hoc: it computes a prototypical rotor from parallel (neutral, transformed) sentence pairs and applies it; it handles one transformation type at a time; it operates at the sentence/discourse level; and it adds no learned per-context projection, no angular address space spanning many relations, and no magnitude axis. SCT is a trained primitive with a learned per-context shadow plane, a compass over a continuum of angular directions, and an explicit magnitude control, addressing concept-level relational navigation rather than the transfer of a single named transformation. RISE establishes that the broad direction is live and independently arrived at; it does not encompass SCT's apparatus.

D.3 Activation-space steering (Angular Steering, Spherical Steering, and related)

A parallel and partly concurrent cluster applies rotation to hidden activations to control model behavior during generation: Angular Steering [13] rotates activations aligned with a target feature for behavioral control, and Spherical Steering [14] applies norm-preserving activation rotation across layers to refine the generation trajectory. These are the works SCT is most at risk of being conflated with, and the distinction is therefore stated explicitly: SCT is *not* an activation-steering or generation-time intervention. It does not rotate hidden states inside a language model's forward pass, does not condition token-probability distributions, and is not autoregressive. It is a standalone, non-autoregressive relational transform that an external system may *call* as a primitive. The "steering" SCT enables is the deliberate querying of a relational geometry, not the rotation of activations during decoding. Conflating the two would both misdescribe SCT and place it, incorrectly, in territory these activation-space methods already occupy.

D.4 Subspace projection and semantic scales

A body of work extracts interpretable structure by projecting embeddings into lower-dimensional subspaces: single-axis "semantic projection" onto a polar-antonym line recovers graded human judgments, and various subspace and relative-representation methods isolate local semantic neighborhoods or align disparate latent spaces by rigid transformation. SCT's shadow plane is a learned 2D instance of this projection idea; single-axis antonym projection is recoverable as the degenerate 1D case of what the shadow plane does in two dimensions. SCT differs in coupling the projection to an explicit rotation operator and a magnitude-scaled lift, so that the subspace is not merely a measurement scale but the stage on which a parameterized transformation is performed and then returned to the full space.

D.5 Adjacent work sharing vocabulary but not mechanism

Several lines of work share the language of "latent navigation" or "trajectories" while operating in a different domain. Embodied and robotic systems (language-to-trajectory guidance, open-vocabulary 3D scene-graph navigation, vision-language-action adapters) and physical-state simulators (molecular latent-space simulators, latent epidemiological-state models) map representations to external physical coordinates or dynamics. These share no mechanism with SCT, which acts on the semantic-relational structure of the embedding itself rather than mapping it to physical motion or state. We note one grey-literature item, a "semantic orbital mechanics" framing that tracks conversational trajectories with a 64-pattern symbolic scheme; the coincidental "64" name-space overlap with SCT's angular-resolution target is noted for completeness, but that framing is observational telemetry over dialogue, mechanistically unrelated to SCT's actuator.

D.6 On the term "shadow plane"

The term "shadow plane" appears in computer graphics and higher-dimensional visualization (shadow mapping, shadow-volume partitioning, projecting higher-dimensional objects onto lower-dimensional control surfaces), not in the semantic-embedding literature. We adopt it because the projection intuition transfers exactly — a shadow is a lower-dimensional, information-lossy cast of a higher-dimensional object — and because the graphics practice of varying a shadow plane's inclination as a length multiplier is a conceptual cousin to SCT's magnitude scaling. The concept has no in-domain precedent under that name; the borrowing is acknowledged here to forestall confusion.

D.7 Summary comparison

Dimension	**SCT**	**RotatE / variants**	**RISE**	**Activation steering (Angular/Spherical)**
Object	Continuous embeddings, any vocabulary	Discrete entity/relation triples	Sentence/discourse embeddings	Hidden activations
Operation	Project to per-context plane → rotate → lift	Element-wise complex rotation	Post-hoc rotor from parallel pairs	Rotate activations in forward pass
Relations addressed	Continuum via compass (0°–360°)	Per-relation learned phase	One transformation type at a time	Per-target-feature direction
Magnitude control	Explicit, continuous (0–1)	None (unit modulus)	None	Implicit / not parameterized
Phase of use	Standalone actuator, called on demand	Training-time link scoring	Offline, post-hoc	Generation-time intervention
Autoregressive	No	No	No	Yes (during decoding)

The pattern across the table is consistent: SCT shares the rotational premise with the left two columns and the embedding-space locus with RISE, but it is alone in combining a per-context learned projection, a navigable angular address space, an explicit magnitude axis, and a non-autoregressive standalone actuator role.

Appendix E: Version History

- **v1.0**: Initial formulation with Ring/Disk/Sphere variants
- **v1.1**: Mathematical bug fix (projection correction)
- **v1.2**: Unified architecture with 2D projection approach
- **v1.3**: Strange Compass Tower metaphor with custom gliders
- **v1.4**: 8-head compass architecture, proximity rewards, 168M parameters
- **v1.5**: Shadow Plane Adapters solving polysemy problem; philosophical foundations section; learning rate hierarchy; expanded loss function with shadow regularization; parameter budget 168M ⟶ 180M
- **v1.6**: Strengthened philosophical foundations with explicit central hypothesis; integrated Hofstadter & Sander with Slotnick evidence; "compressed latent computation" framing with clear attribution; Hofstadter's urban concept-growth metaphor; explicit gap analysis; Cognitive Science Evidence Map (Appendix C)
- **v1.7**: Math audit pass. Parameter budget recomputed from architecture and corrected: embedding 92.16M, FFN 28.31M (both matrices per layer), attention 10.62M (uneven 48/48/96 split), LayerNorm/biases ~57K, positional encoding ~25K, geometric encoders ~13K, shadow basis 18.4K, alignment 12 (B) or 48 (A); total ~131.2M (previously ~180M). Pristine basis vectors removed throughout as architecturally unused: Sections 3.2, 3.6, 4.1, 4.2, 4.3, 5.2, 6.1, 6.3, 7.1, and Footnote 2 updated accordingly. Appendix A.1 involution proof corrected (sign error fixed) and restated precisely: exact at $m \in \{0, 1\}$, approximate with residual $4m(1\text{-}m)\cdot\|proj\_shadow(c)\|$ for $m \in (0, 1)$. Property 2 restated to match. Property 3 restated to distinguish context-specific from shared parameters. Section 3.5 angular proximity function corrected to handle 0°/360° seam via modular distance. Section 4.4 L_geometric split to enforce involution only at $m = 1$ (avoiding self-defeating training signal); loss function notation flattened to one lambda per term. Appendix A.2 expanded with Variant A invertibility proof via SVD parameterization.
- **v1.8**: Added explicit statement of methodological commitment distinguishing the terminal goal (realization of an SCT as a queryable analogical-navigation primitive) from the instrumental architecture (this paper's leading but revisable design); inserted at the close of Section 2.1 and echoed in the Conclusion. No changes to mathematics, parameter budget, or architecture specification.
- **v1.9**: Presentation, mathematical-honesty, experimental-scaffolding, and prior-art audit pass. Removed the gauge-redundant SO(2) alignment angles, which are inert under abelian conjugation ($W_x^{-1}R(\theta)W_x = R(\theta)$) and therefore dead parameters; Section 3.4 restructured from Variant

A / Variant B into two genuine configurations — Shadow-direct (no alignment, rotation in shadow coordinates) and Shadow-affine (SVD-parameterized GL(2) alignment, rotation in a rotation plane) — with the SO(2)-collapse rationale stated explicitly as Section 3.4.3. “Pristine plane / coordinates” renamed to “rotation plane / coordinates” throughout (Sections 3.2, 3.3, 3.6, 4.1, 4.4, 7.1, Footnote 1, Appendix A). Section 3.3 forward pipeline rewritten as a single conditional sequence (Steps 2 and 4 Shadow-affine-only). Property 2 and Appendix A.1 reframed as an architectural inductive bias rather than a learned property; Appendix A.2 reduced to the Shadow-affine invertibility proof (the inert orthogonal proof removed). MLP baseline added and Section 6.3 reorganized into a four-rung ablation ladder (MLP → no-shadow-plane → Shadow-direct → Shadow-affine) plus a compass-head-count ablation; Falsifiability Criteria added as Section 6.5. Agentic-routing use case acknowledged at the close of Section 1, worded to preclude an activation-steering misreading. Engineered-approximation limitations note added at the end of Section 2.8, with a pointer from the Conclusion. Prior- and concurrent-art summary added as Section 2.9, with a detailed analysis in the new Appendix D (RotatE and descendants, RISE [arXiv:2510.09790], the activation-steering cluster, subspace/semantic-projection methods, and adjacent-domain work); references [12]–[14] added; independent and concurrent development relative to RISE and the rotation-steering cluster acknowledged as of June 2026. Architecture and mathematical content otherwise unchanged from v1.7; parameter budget reduced by 12 inert alignment parameters (immaterial at ~131.2M). Prior Version History relettered from Appendix D to Appendix E to accommodate the new Appendix D.

- **v1.10**: Pre-publication cleanup. (1) Completed reference [14] (You, Deng & Chen, Spherical Steering, arXiv:2602.08169, ICML 2026). (2) Inserted pagebreak markers ahead of References and each appendix for clean PDF pagination. No changes to architecture, forward transformation, proofs, or parameter budget.